Panskura Banamali College (Autonomous)
B.Sc.(H) 1st Semester Examination 2022 (Online mode)
Subject : BCA **Paper: CC2P**
Paper Name : Computer Fundamental with Digital Electronics
F.M. : 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2021UG009967, paper_code : CC2P , then file name will be :: 2021UG009967- CC2P.pdf for question 'A' and 2021UG009967- CC2P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. Draw a block diagram for 4 full adders processing the addition on 4 bits of A and 4 bits of B register. Show the different implementation of half subtractor and full subtractor. 6+6
2. Design a full adder using any universal logic gates & verify its truth table.
3. Implement Boolean function F(A,B,C,D)=Σ m(0,1,3,4,8,9,15) using 8:1 multiplexer. Design 3-bit binary counter using T flip-flop. 8+4
4. Design a combinational circuit that accepts a three-bit number and generates an output binary number equal to the square of the input number.
5. Draw the logic diagram of clocked RS Flip-Flop and explain its operation. Design a counter with the following binary sequence: 0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use RS flip flops. 6+6
6. Draw a truth table for a 4 x 1 line multiplexer without enable line.
   Draw a truth table for a 4 x 1 line multiplexer with enable line having active LOW. 6+6
7. Draw the K-map and Implement the simplified expression using basic gates. (8)∑Develop the given function Y (M, N, O, P, Q) = (31,29,25,23,21,13,9,6,4,2,0) .Implement F= A'B'D'+ B'C'+E' using NAND gates. 8+4
8. Briefly describe about Parallel-Access Shift Register and Three-Bit Up-Counter . 6+6

Marks distribution:

a. Problem Definition -	1 Marks
b. Algorithm/Flow Chart/ Pseudo-code–	5 Marks
c. Program listing -	3 Marks
d. Output/Result with discussion-	3 Marks

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What are the basic gates in MOS logic family?
2. Given the Boolean function: F=xy+x'y'+y'z Implement it with only OR and NOT gates.
3. Simplify the following Boolean expressions by manipulation of Boolean algebra. F(x,y,z)=xy+xyz+xyz'+x'yz
4. Simplify the following Boolean expressions by manipulation of Boolean algebra. F(A,B,C,D)=A'C(A'BD)'+A'BC'D'+AB'C
5. What is Combinational circuit ?
6. What is MAR ?
7. Describe Parallel Subtractor .
8. Define Priority Encoder .
9. Implement NOT gate using 2 : 1 Mux.
10. What is Flip Flop ?

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